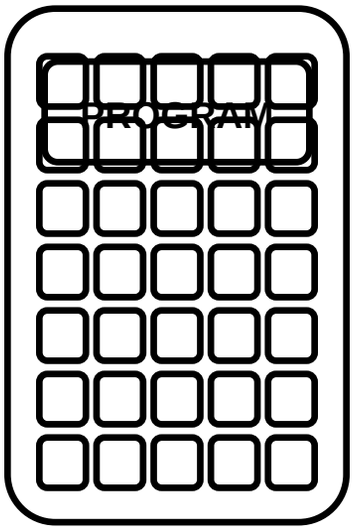
CASIO fx-50FH II



S5 PROGRAM LIBRARY

中五計算機程式． 精選版

16 tested programs · 7 memory-safe packs · 680-byte aware

Program 01 一元二次方程工具
Quadratic X-Ray

Program 02 等差及等比數列
Sequence Engine

Program 03 坐標幾何及圓
Coordinate Lab

Program 04 三角方程解
Trig 360

Program 05 聯立二元一次方程
Simultaneous 2×2 Solver

Program 06 三點定圓
Circle Through Three Points

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Program 09 直線與圓的參數 k 交點
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Program 10 數列目標項數
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Program 11 一元三次方程完整求根
Complete Cubic Solver

Program 12 不放回抽樣概率
Without-Replacement Probability

Program 13 變換根方程
Transformed-Roots Builder

Program 14 圓外一點的切線
Tangents from a Point to a Circle

Program 15 年金與貸款
Annuity and Loan Engine

Program 16 任意三角形四心
Four Centres of Any Triangle

Built from the canonical tested web catalog · Revised 13 August 2026

PRGM Command Reference

程式模式按鍵速查

Command	Physical keys	Command	Physical keys
?	SHIFT 3 1	ClrMemory	SHIFT 9 1
: / new line	SHIFT 3 3 or EXE	ClrStat	SHIFT 9 1
◢ output	SHIFT 3 4	Goto	SHIFT 3 ▸ ▸ ▸ 1
→ assign	SHIFT 3 2	Lbl	SHIFT 3 ▸ ▸ ▸ 2
⇒	SHIFT 3 ▸ 1	If	SHIFT 3 ◂ 1
=	SHIFT 3 ▸ 2	Then	SHIFT 3 ◂ 2
≠	SHIFT 3 ▸ 3	Else	SHIFT 3 ◂ ◂ 1
> / <	SHIFT 3 ▸ ▸ 1 / 2	IfEnd	SHIFT 3 ◂ ◂ 1
≥ / ≤	SHIFT 3 ▸ ▸ 3 / 4	While	SHIFT 3 ▸ ▸ ▸ 1
A B C D	ALPHA + letter key	WhileEnd	SHIFT 3 ▸ ▸ ▸ 2
X Y M	ALPHA +) , M+	For / To / Step	SHIFT 3 ◂ ◂ ◂ 1/2/3
Rnd(	SHIFT 0	Next	SHIFT 3 ◂ ◂ ◂ ◂ 1
Fix / Sci / Norm	SHIFT MODE ▸ 1/2/3	Abs(	SHIFT)
Pol(/ Rec(	SHIFT + / SHIFT -	DT	M+ in SD/REG

Tip: an ordinary minus sign is accepted in Program Studio; use the physical (-) key for negative input on the calculator.

Program 1

Quadratic X-Ray

一元二次方程工具

用法

一次輸入係數，即時計算判別式、頂點及所有實根。

INPUT A=a，B=b，C=c

MODE COMP · 80 bytes

OUTPUT 判別式、頂點 (h,k)、實根

Program steps

```
?→A
?→B
?→C
B²-4AC→D
D◢-B÷2÷A◢-D÷4÷A◢D<0⇒Goto 9
(√(D)-B)÷2÷A→C
C◢D=0⇒Goto 9
-B÷A-C◢Lbl 9
```

示範 Example

```
TYPE  A=1, B=-5, C=6
GET   Δ=1, h=2.5, k=-0.25, roots 3 and 2
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 2

Sequence Engine

等差及等比數列

用法

處理等差、有限等比及無限等比數列的項及和。

INPUT M=類型；A=首項；B=公差/公比；C=n

MODE COMP · 110 bytes

OUTPUT 指定項及部分和/無限和

Program steps

```
?→M
?→A
?→B
M=3⇒A÷(1-B)→X
M=3⇒Goto 9
?→C
M=1⇒A+(C-1)B→D
M=2⇒AB^(C-1)→D
D◢M=1⇒C(A+D)÷2→X
M=2⇒(A-BD)÷(1-B)→X
Lbl 9
X◢
```

示範 Example

```
TYPE  M=1, A=2, B=3, C=5
GET   T₅=14, S₅=40
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 3

Coordinate Lab

坐標幾何及圓

用法

計算兩點距離、中點、斜率，或由一般式求圓心及半徑。

INPUT M=1 兩點；M=2 圓的一般式係數

MODE COMP · 104 bytes

OUTPUT 距離、中點、斜率 / 圓心、半徑

Program steps

```
?→M
?→A
?→B
?→C
M=2⇒Goto 2
?→D
C-A→X
D-B→Y
√(X²+Y²)◢(A+C)÷2◢(B+D)÷2◢Y÷X◢Goto 9
Lbl 2
-A÷2◢-B÷2◢√(A²+B²-4C)÷2◢Lbl 9
```

示範 Example

```
TYPE  M=1, A=0, B=0, C=3, D=4
GET   distance 5, midpoint (1.5,2), slope 1.333333333
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 4

Trig 360

三角方程解

用法

求 0° 至 360° 內所有 sin、cos 或 tan 方程的解。

INPUT M=1/2/3；D=已分離的三角函數值

MODE COMP · 184 bytes

OUTPUT 區間內所有不同角度

Program steps

```
Deg
?→M
?→D
M=1⇒Goto 1
M=2⇒Goto 2
tan^-1(D)→X
X<0⇒X+180→X
X◢X+180◢X≠0⇒Goto 9
360◢Goto 9
Lbl 1
sin^-1(D)→X
X<0⇒Goto 4
X◢X=90⇒Goto 9
180-X◢X≠0⇒Goto 9
360◢Goto 9
Lbl 4
180-X◢X=-90⇒Goto 9
360+X◢Goto 9
Lbl 2
cos^-1(D)→X
X◢X=180⇒Goto 9
360-X◢Lbl 9
```

示範 Example

2sin θ=1 → TYPE M=1, D=0.5 → GET 30°, 150°
cos θ=1 → TYPE M=2, D=1 → GET 0°, 360°
tan θ=-1 → TYPE M=3, D=-1 → GET 135°, 315°

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 5

Simultaneous 2×2 Solver

聯立二元一次方程

用法

輸入兩條直線方程的六個係數，求唯一交點。

INPUT A,B,C 及 D,X,Y 為兩式係數

MODE COMP · 52 bytes

OUTPUT x、y

Program steps

```
?→A
?→B
?→C
?→D
?→X
?→Y
AX-BD→M
(CX-BY)÷M◢(AY-CD)÷M◢
```

示範 Example

```
TYPE  A=2, B=1, C=7, D=1, X=-1, Y=2
GET   x=3, y=1
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 6

Circle Through Three Points

三點定圓

用法

由三個不共線點求圓心、半徑及一般式。

INPUT A,B；C,D；X,Y 為三點坐標

OUTPUT 圓心、半徑、一般式係數

MODE COMP · 129 bytes

Program steps

```
?→A
?→B
?→C
?→D
?→X
?→Y
C-A→C
D-B→D
X-A→X
Y-B→Y
(C(X²+Y²)-X(C²+D²))÷2÷(CY-DX)→M
((C²+D²)Y-(X²+Y²)D)÷2÷(CY-DX)→C
A+C◢B+M◢√(C²+M²)◢
```

示範 Example

```
TYPE  A=0, B=0, C=4, D=0, X=0, Y=6
GET   centre (2,3), radius 3.605551275
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 7

Growth and Decay Solver

增長與衰減

用法

在複式增長或衰減模型中求終值、時間、率或初值。

INPUT M=未知量類型；其餘輸入已知數值 MODE COMP · 110 bytes

OUTPUT 所選未知量

Program steps

```
?→M
?→A
?→B
?→C
M=1⇒A(1+B÷100)^(C)→X
M=2⇒log(C÷A)÷log(1+B÷100)→X
M=3⇒100((B÷A)^(1÷C)-1)→X
M=4⇒A÷(1+B÷100)^(C)→X
X◢
```

示範 Example

```
TYPE  M=1, A=1000, B=5, C=3     → GET 1157.625
TYPE  M=2, A=1000, B=10, C=1210 → GET 2 periods
TYPE  M=3, A=1000, B=1210, C=2  → GET 10%
TYPE  M=4, A=1210, B=10, C=2    → GET 1000
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 8

Combined Population Statistics

合併統計數據

用法

只有兩組人數、平均數及母體標準差時，合併兩組數據。

INPUT A,B,C 及 D,X,Y 為兩組 n, μ, σ

MODE COMP · 73 bytes

OUTPUT 合併平均數及母體標準差

Program steps

```
?→A
?→B
?→C
?→D
?→X
?→Y
(AB+DX)÷(A+D)◢√((A(C²+B²)+D(Y²+X²))÷(A+D)-(Ans)²)◢
```

示範 Example

TYPE A=2, B=10, C=2, D=2, X=14, Y=2
GET combined mean 12, combined SD 2.828427125

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 9

Line–Circle Intersections

直線與圓的參數 k 交點

用法

坐標可為數字、k 或 ak+b；同時求 k 及所有交點。

INPUT 圓 A,B,C；x=D+Xk；y=Y+Mk

MODE COMP · 140 bytes

OUTPUT 每組 k、x、y；切線只輸出一組

Program steps

```
?→A
?→B
?→C
?→D
?→X
?→Y
?→M
D²+Y²+AD+BY+C→C
2(DX+YM)+AX+BM→B
X²+M²→A
B²-4AC→C
(√(C)-B)÷(2A)→C
C◢D+XC◢Y+MC◢
-B÷A-C→B
B=C⇒Goto 9
B◢D+XB◢Y+MB◢
Lbl 9
```

示範 Example

```
TYPE  A=-4, B=-2, C=-4, D=0, X=1, Y=1, M=0
GET   k=5, point (5,1); k=-1, point (-1,1)
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 10

Sequence Target Solver

數列目標項數

用法

求部分和首次達到指定目標時的最小正整數 n。

INPUT M=AP/GP；A=首項；B=公差/公比；C=目標

MODE COMP · 72 bytes

OUTPUT 最小 n 及對應 S□

Program steps

```
?→M
?→A
?→B
?→C
0→X
0→Y
A→D
While Y<C
X+1→X
Y+D→Y
M=1⇒D+B→D
M=2⇒DB→D
WhileEnd
X◢Y◢
```

示範 Example

TYPE M=1, A=3, B=2, C=100
GET n=10, S_{10}=120

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 11

Complete Cubic Solver

一元三次方程完整求根

用法

毋須先知道因式，直接求一元三次方程的所有不同實根。

INPUT A,B,C,M 為 $Ax^3+Bx^2+Cx+D=0$ 的係數

MODE COMP · 273 bytes

OUTPUT 一個、兩個或三個不同實根

Program steps

```
?→A
?→B
?→C
?→M
2×B³÷(27×A³)-B×C÷(3×A²)+M÷A→D
C÷A-B²÷(3×A²)→X
B÷(3×A)→Y
(D÷2)²+(X÷3)³→M
M<0⇒Goto 1
M=0⇒Goto 2
∛((-D÷2+√(M)))+∛((-D÷2-√(M)))-Y◢
Goto 9
Lbl 2
∛((-D÷2))→A
2×A-Y◢
A=0⇒Goto 9
-A-Y◢
Goto 9
Lbl 1
Deg
cos^-1((3×D÷(2×X))×√((-3÷X)))÷3→M
2×√((-X÷3))→A
A×cos(M)-Y◢
A×cos(M - 120)-Y◢
A×cos(M - 240)-Y◢
Lbl 9
```

示範 Example

```
TYPE  A=1, B=6, C=11, M=6
GET   roots -1, -2, -3
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 12

Without-Replacement Probability

不放回抽樣概率

用法

以組合公式計算恰好、至少或至多指定數目的概率。

INPUT M=1/2/3；A,B=兩類數量；C=抽取數；D=目標

MODE COMP · 130 bytes

OUTPUT 所需概率

Program steps

```
?→M
?→A
?→B
?→C
?→D
0→X
C-B>0⇒C-B→X
C→Y
A<C⇒A→Y
M=1⇒D→X
M=1⇒D→Y
M=2⇒D>X⇒D→X
M=3⇒D<Y⇒D→Y
0→M
While X≤Y
A nCr X B nCr(C-X)÷(A+B)nCr C M+
X+1→X
WhileEnd
M◢
```

示範 Example

```
TYPE  M=1, A=5, B=4, C=2, D=2  → GET P(exactly 2 red)=0.2777777778 = 5/18
TYPE  M=2, A=5, B=4, C=2, D=1  → GET P(at least 1 red)=0.8333333333 = 5/6
TYPE  M=3, A=5, B=4, C=2, D=1  → GET P(at most 1 red)=0.7222222222 = 13/18
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 13

Transformed-Roots Builder

變換根方程

用法

不先求原根，直接建立線性變換根或平方根的新二次方程。

INPUT M=模式；A,B,C=原式；模式 1 再輸入 X,Y

MODE COMP · 83 bytes

OUTPUT 新首一二次方程的 P、Q

Program steps

```
?→M
?→A
?→B
?→C
M=2⇒Goto 2
?→X
?→Y
XB÷A-2Y◢X²C÷A-YAns-Y²◢Goto 9
Lbl 2
C÷A→X
2X-(B÷A)²◢X²◢Lbl 9
```

示範 Example

TYPE M=1, A=1, B=-5, C=6, X=2, Y=1
GET P=-12, Q=35
WRITE z²-12z+35=0

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 14

Tangents from a Point to a Circle

圓外一點的切線

用法

由圓心、半徑及圓外點求切線長及兩個切點。

INPUT A,B=圓心；C=半徑；D,X=圓外點

OUTPUT 切線長及兩個切點

MODE COMP · 104 bytes

Program steps

```
?→A
?→B
?→C
?→D
?→X
D-A→D
X-B→X
D²+X²→M
√((M)-C²)→Y
Y◢C÷M→M
A+M(CD-XY)◢B+M(CX+DY)◢A+M(CD+XY)◢B+M(CX-DY)◢
```

示範 Example

```
TYPE  A=0, B=0, C=5, D=13, X=0
GET   tangent length 12
      contact points (1.923076923,4.615384615)
                     (1.923076923,-4.615384615)
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 15

Annuity and Loan Engine

年金與貸款

用法

處理期末供款的儲蓄終值、貸款供款及尚欠餘額。

INPUT M=模式；A=本金；B=每期利率%；C=期數；D=供款

MODE COMP · 107 bytes

OUTPUT 所選金融數值

Program steps

```
?→M
?→A
?→B
B÷100→B
?→C
(1+B)^(C)→X
M=2⇒AB÷(1-1÷X)→Y
M=2⇒Goto 9
?→D
M=1⇒AX+D(X-1)÷B→Y
M=3⇒AX-D(X-1)÷B→Y
Lbl 9
Y◢
```

示範 Example

```
TYPE  M=1, A=1000, B=1, C=12, D=100
GET   future value 2395.075331
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 16

Four Centres of Any Triangle

任意三角形四心

用法

輸入任意三個不共線頂點，求形心、內心、垂心及外心。

INPUT REG 模式；A,B；C,D；X,Y 為三點坐標

MODE REG · 385 bytes

OUTPUT G、I、H、O 四組坐標

Program steps

```
ClrStat
?→A
?→B
A,B DT
?→C
?→D
C,D DT
?→X
?→Y
X,Y DT
Σx÷3◢
Σy÷3◢
√((C-X)²+(D-Y)²)+√((X-A)²+(Y-B)²)+√((A-C)²+(B-D)²)→M
(√((C-X)²+(D-Y)²)×A+√((X-A)²+(Y-B)²)×C+√((A-C)²+(B-D)²)×X)÷M◢
(√((C-X)²+(D-Y)²)×B+√((X-A)²+(Y-B)²)×D+√((A-C)²+(B-D)²)×Y)÷M◢
2×(A×(D-Y)+C×(Y-B)+X×(B-D))→M
Pol(((A²+B²)×(D-Y)+(C²+D²)×(Y-B)+(X²+Y²)×(B-D))÷M,((A²+B²)×(X-C)+(C²+D²)×(A-X)+(X²+Y²)×(C-A))÷M)
Rec(X,Y)
Σx-2X◢
Σy-2Y◢
X◢
Y◢
```

示範 Example

```
TYPE  A=0, B=0, C=24, D=0, X=12, Y=9
GET   G=(12,3), I=(12,4), H=(12,16), O=(12,-3.5)
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.